

PROBLEMS

⊕ Problem available in WileyPLUS at instructor's discretion.

Section 4.2 Node Voltage Analysis of Circuits with Current Sources

P 4.2-1 ⊕ The node voltages in the circuit of Figure P 4.2-1 are $v_1 = -4$ V and $v_2 = 2$ V. Determine i , the current of the current source.

Answer: $i = 1.5$ A

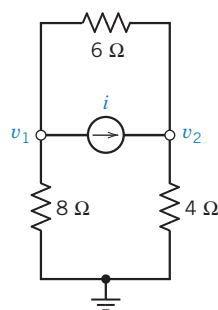


Figure P 4.2-1

P 4.2-2 Determine the node voltages for the circuit of Figure P 4.2-2.

Answer: $v_1 = 2$ V, $v_2 = 30$ V, and $v_3 = 24$ V

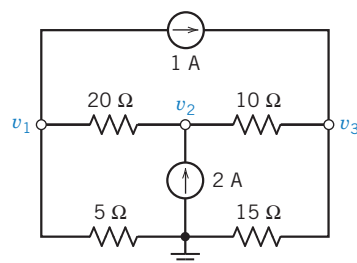


Figure P 4.2-2

P 4.2-3 The encircled numbers in the circuit shown in Figure P 4.2-3 are node numbers. Determine the values of the corresponding node voltages v_1 and v_2 .

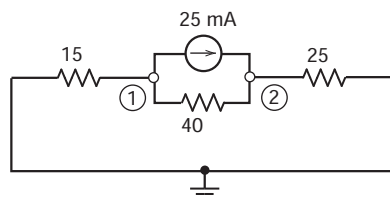


Figure P 4.2-3

P 4.2-4 ⊕ Consider the circuit shown in Figure P 4.2-4. Find values of the resistances R_1 and R_2 that cause the voltages v_1 and v_2 to be $v_1 = 1$ V and $v_2 = 2$ V.

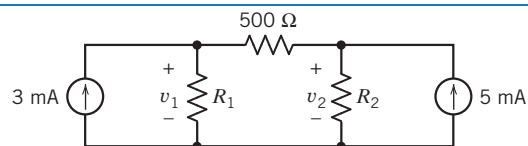


Figure P 4.2-4

P 4.2-5 Find the voltage v for the circuit shown in Figure P 4.2-5.

Answer: $v = 21.7$ mV

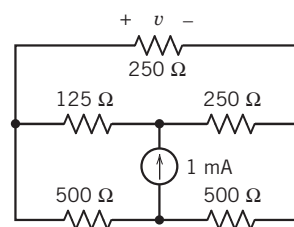


Figure P 4.2-5

P 4.2-6 Simplify the circuit shown in Figure P 4.2-6 by replacing series and parallel resistors with equivalent resistors; then analyze the simplified circuit by writing and solving node equations. (a) Determine the power supplied by each current source. (b) Determine the power received by the 12-Ω resistor.

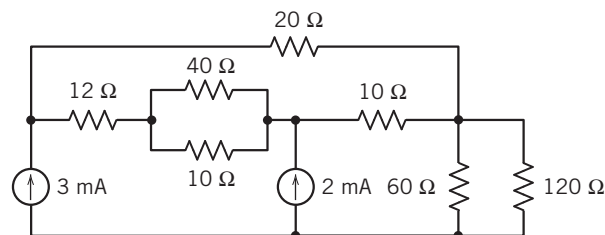


Figure P 4.2-6

P 4.2-7 The node voltages in the circuit shown in Figure P 4.2-7 are $v_a = 7$ V and $v_b = 10$ V. Determine values of the current source current, i_s , and the resistance, R .

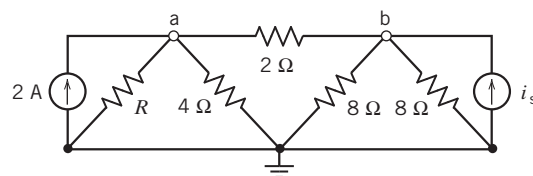


Figure P 4.2-7

P 4.2-8 The encircled numbers in the circuit shown in Figure P 4.2-8 are node numbers. The corresponding node voltages are v_1 and v_2 . The node equation representing this circuit is

$$\begin{bmatrix} 0.225 & -0.125 \\ -0.125 & 0.125 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} -3 \\ 2 \end{bmatrix}$$

- (a) Determine the values of R and I_s in Figure P 4.2-8.
 (b) Determine the value of the power supplied by the 3-A current source.

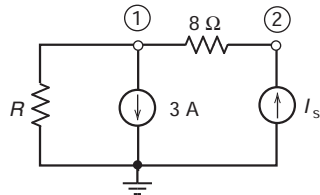


Figure P 4.2-8

Section 4.3 Node Voltage Analysis of Circuits with Current and Voltage Sources

P 4.3-1 + The voltmeter in Figure P 4.3-1 measures v_c , the node voltage at node c. Determine the value of v_c .

Answer: $v_c = 2$ V

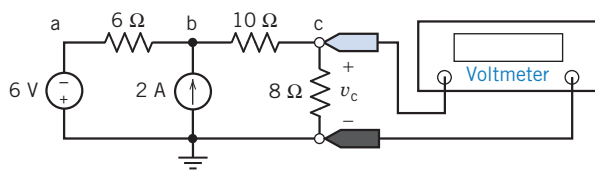


Figure P 4.3-1

P 4.3-2 + The voltages v_a , v_b , v_c , and v_d in Figure P 4.3-2 are the node voltages corresponding to nodes a, b, c, and d. The current i is the current in a short circuit connected between nodes b and c. Determine the values of v_a , v_b , v_c , and v_d and of i .

Answer: $v_a = -12$ V, $v_b = v_c = 4$ V, $v_d = -4$ V, $i = 2$ mA

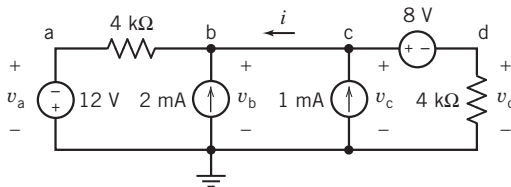


Figure P 4.3-2

P 4.3-3 Determine the values of the power supplied by each of the sources in the circuit shown in Figure P 4.3-3.

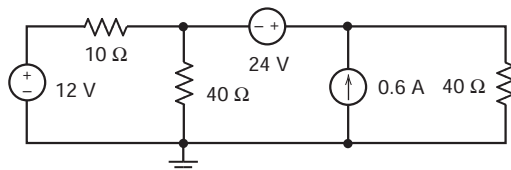


Figure P 4.3-3

P 4.3-4 Determine the values of the node voltages v_1 , v_2 , and v_3 in the circuit shown in Figure P 4.3-4.

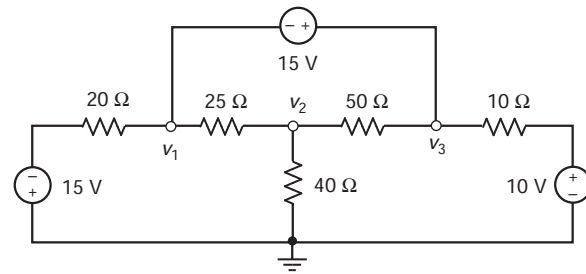


Figure P 4.3-4

P 4.3-5 + The voltages v_a , v_b , and v_c in Figure P 4.3-5 are the node voltages corresponding to nodes a, b, and c. The values of these voltages are:

$$v_a = 12 \text{ V}, v_b = 9.882 \text{ V}, \text{ and } v_c = 5.294 \text{ V}$$

Determine the power supplied by the voltage source.

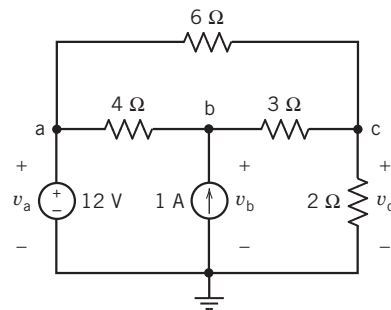


Figure P 4.3-5

P 4.3-6 + The voltmeter in the circuit of Figure P 4.3-6 measures a node voltage. The value of that node voltage depends on the value of the resistance R .

- (a) Determine the value of the resistance R that will cause the voltage measured by the voltmeter to be 4 V.
 (b) Determine the voltage measured by the voltmeter when $R = 1.2 \text{ k}\Omega = 1200 \Omega$.

Answers: (a) 6 kΩ (b) 2 V

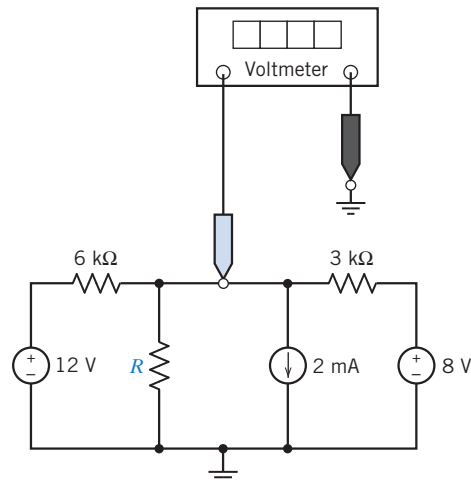


Figure P 4.3-6

P 4.3-7 $\oplus$ Determine the values of the node voltages v_1 and v_2 in Figure P 4.3-7. Determine the values of the currents i_a and i_b .

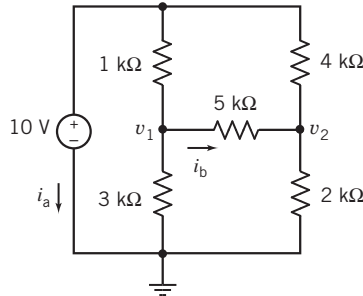


Figure P 4.3-7

P 4.3-8 The circuit shown in Figure P 4.3-8 has two inputs, v_1 and v_2 , and one output, v_o . The output is related to the input by the equation

$$v_o = av_1 + bv_2$$

where a and b are constants that depend on R_1 , R_2 , and R_3 .

- Determine the values of the coefficients a and b when $R_1 = 10\ \Omega$, $R_2 = 40\ \Omega$, and $R_3 = 8\ \Omega$.
- Determine the values of the coefficients a and b when $R_1 = R_2$ and $R_3 = R_1 \parallel R_2$.

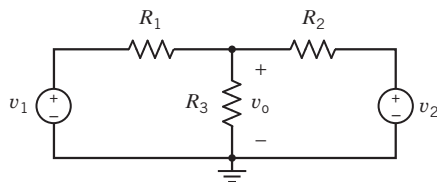


Figure P 4.3-8

P 4.3-9 Determine the values of the node voltages of the circuit shown in Figure P 4.3-9.

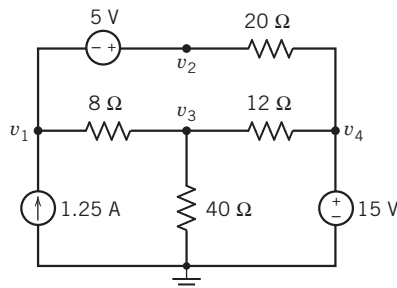


Figure P 4.3-9

P 4.3-10 Figure P 4.3-10 shows a measurement made in the laboratory. Your lab partner forgot to record the values of R_1 , R_2 , and R_3 . He thinks that the two resistors were 10-k Ω resistors

and the other was a 5-k Ω resistor. Is this possible? Which resistor is the 5-k Ω resistor?

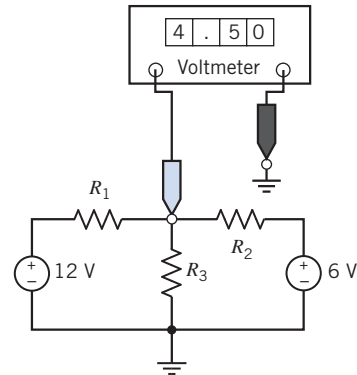


Figure P 4.3-10

P 4.3-11 Determine the values of the power supplied by each of the sources in the circuit shown in Figure P 4.3-11.

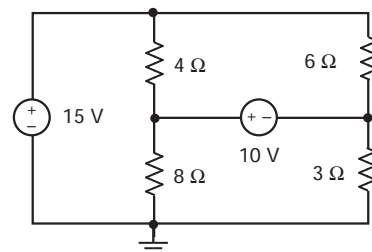


Figure P 4.3-11

P 4.3-12 Determine the values of the node voltages of the circuit shown in Figure P 4.3-12.

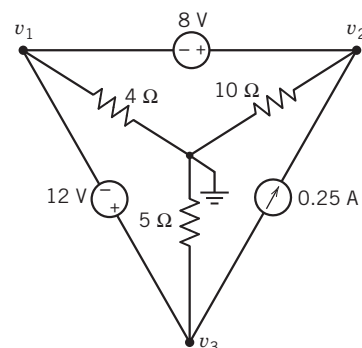


Figure P 4.3-12

P 4.3-13 $\oplus$ Determine the values of node voltages v_1 and v_2 in the circuit shown in Figure P 4.3-13.

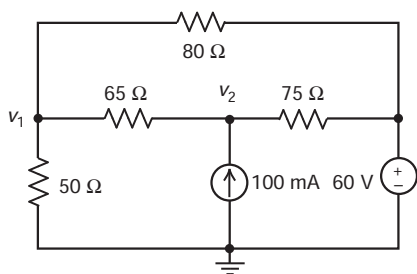


Figure P 4.3-13

P 4.3-14 The voltage source in the circuit shown in Figure P 4.3-14 supplies 83.802 W. The current source supplies 17.572 W. Determine the values of the node voltages v_1 and v_2 .

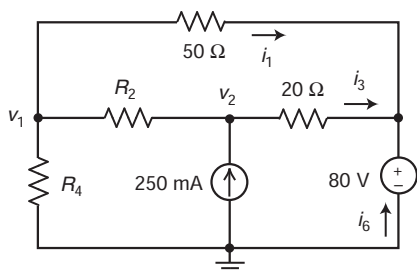


Figure P 4.3-14

Section 4.4 Node Voltage Analysis with Dependent Sources

P 4.4-1 The voltages v_a , v_b , and v_c in Figure P 4.4-1 are the node voltages corresponding to nodes a, b, and c. The values of these voltages are:

$$v_a = 8.667 \text{ V}, v_b = 2 \text{ V}, \text{ and } v_c = 10 \text{ V}$$

Determine the value of A , the gain of the dependent source.

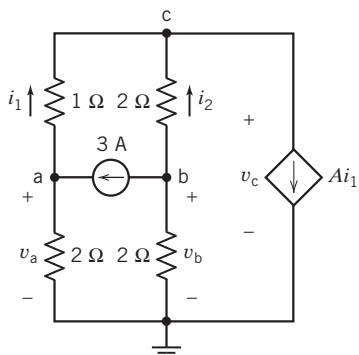


Figure P 4.4-1

P 4.4-2 Find i_b for the circuit shown in Figure P 4.4-2.

Answer: $i_b = -12 \text{ mA}$

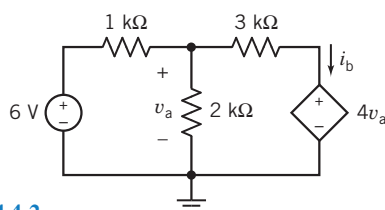


Figure P 4.4-2

P 4.4-3 Determine the node voltage v_b for the circuit of Figure P 4.4-3.

Answer: $v_b = 1.5 \text{ V}$

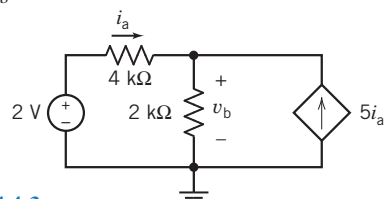


Figure P 4.4-3

P 4.4-4 The circled numbers in Figure P 4.4-4 are node numbers. The node voltages of this circuit are $v_1 = 10 \text{ V}$, $v_2 = 14 \text{ V}$, and $v_3 = 12 \text{ V}$.

(a) Determine the value of the current i_b .

(b) Determine the value of r , the gain of the CCVS.

Answers: (a) -2 A (b) 4 V/A

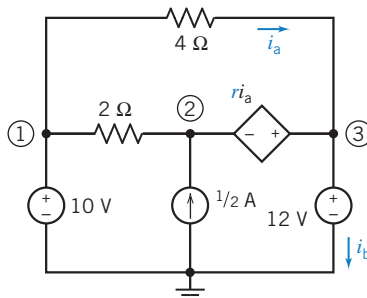


Figure P 4.4-4

P 4.4-5 Determine the value of the current i_x in the circuit of Figure P 4.4-5.

Answer: $i_x = 2.4 \text{ A}$

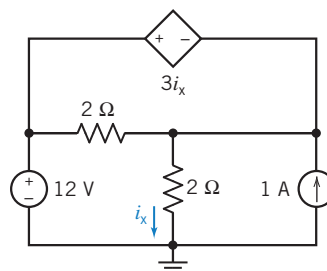


Figure P 4.4-5

P 4.4-6 The encircled numbers in the circuit shown in Figure P 4.4-6 are node numbers. Determine the value of the power supplied by the CCVS.

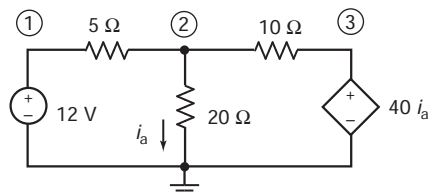


Figure P 4.4-6

P 4.4-7 The encircled numbers in the circuit shown in Figure P 4.4-7 are node numbers. The corresponding node voltages are:

$$v_1 = 9.74 \text{ V and } v_2 = 6.09 \text{ V}$$

Determine the values of the gains of the dependent sources, r and g .

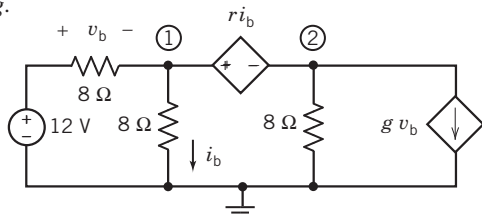


Figure P 4.4-7

P 4.4-8 Determine the value of the power supplied by the dependent source in Figure P 4.4-8.

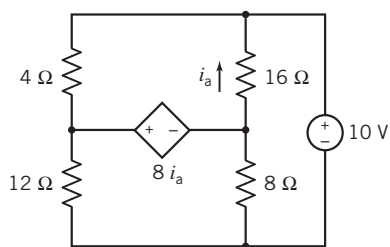


Figure P 4.4-8

P 4.4-9 The node voltages in the circuit shown in Figure P 4.4-9 are

$$v_1 = 4 \text{ V, } v_2 = 0 \text{ V, and } v_3 = -6 \text{ V}$$

Determine the values of the resistance R and of the gain b of the CCCS.

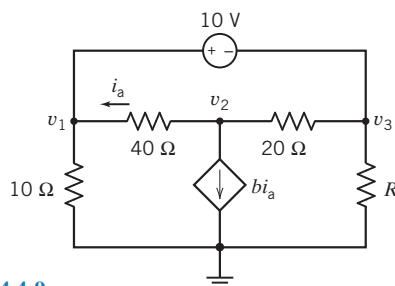


Figure P 4.4-9

P 4.4-10 The value of the node voltage at node b in the circuit shown in Figure P 4.4-10 is $v_b = 18 \text{ V}$.

- (a) Determine the value of A , the gain of the dependent source.
- (b) Determine the power supplied by the dependent source.

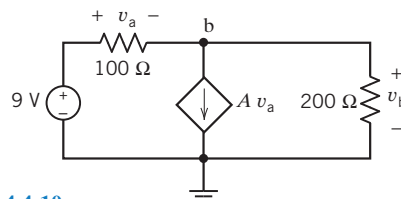


Figure P 4.4-10

***P 4.4-11** Determine the power supplied by the dependent source in the circuit shown in Figure P 4.4-11.

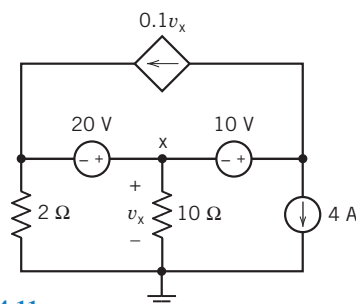


Figure P 4.4-11

***P 4.4-12** Determine values of the node voltages v_1 , v_2 , v_3 , v_4 , and v_5 in the circuit shown in Figure P 4.4-12.

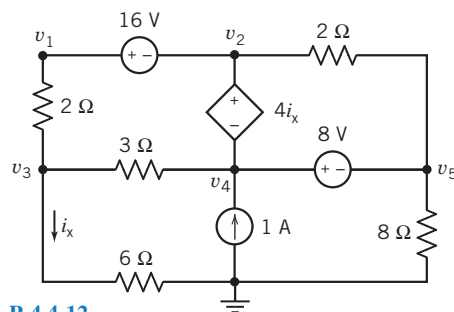


Figure P 4.4-12

***P 4.4-13** Determine values of the node voltages v_1 , v_2 , v_3 , v_4 , and v_5 in the circuit shown in Figure P 4.4-13.

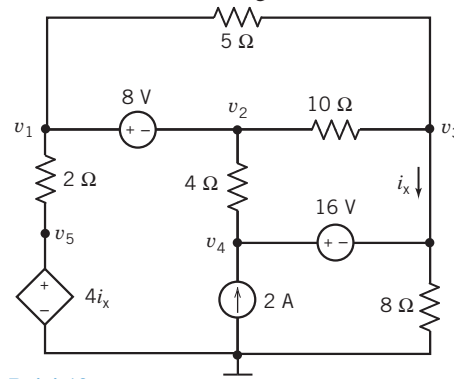


Figure P 4.4-13

P 4.4-14 The voltages v_1 , v_2 , v_3 , and v_4 are the node voltages corresponding to nodes 1, 2, 3, and 4 in Figure P 4.4-14. Determine the values of these node voltages.

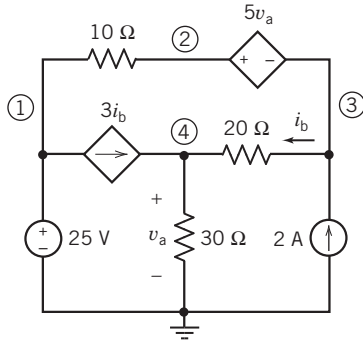


Figure P 4.4-14

P 4.4-15 The voltages v_1 , v_2 , v_3 , and v_4 in Figure P 4.4-15 are the node voltages corresponding to nodes 1, 2, 3, and 4. The values of these voltages are

$$v_1 = 10 \text{ V}, v_2 = 75 \text{ V}, v_3 = -15 \text{ V}, \text{ and } v_4 = 22.5 \text{ V}$$

Determine the values of the gains of the dependent sources, A and B , and of the resistance R_1 .

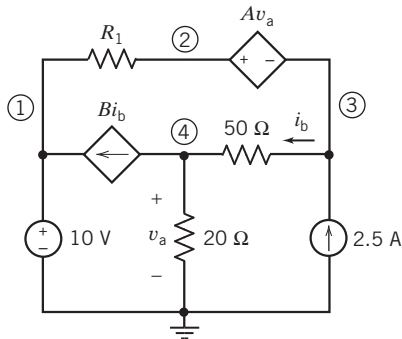


Figure P 4.4-15

P 4.4-16 The voltages v_1 , v_2 , and v_3 in Figure P 4.4-16 are the node voltages corresponding to nodes 1, 2, and 3. The values of these voltages are

$$v_1 = 12 \text{ V}, v_2 = 21 \text{ V}, \text{ and } v_3 = -3 \text{ V}$$

- Determine the values of the resistances R_1 and R_2 .
- Determine the power supplied by each source.

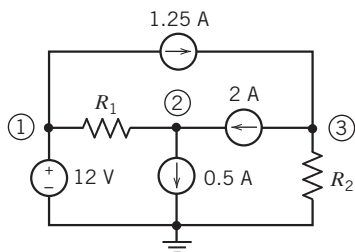


Figure P 4.4-16

P 4.4-17 The voltages v_1 , v_2 , and v_3 in Figure P 4.4-17 are the node voltages corresponding to nodes 1, 2, and 3. The values of these voltages are

$$v_1 = 12 \text{ V}, v_2 = 9.6 \text{ V}, \text{ and } v_3 = -1.33 \text{ V}$$

- Determine the values of the resistances R_1 and R_2 .
- Determine the power supplied by each source.

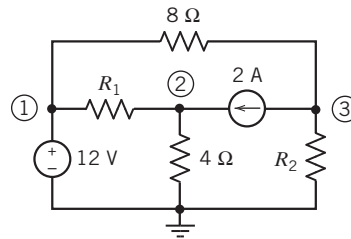


Figure P 4.4-17

P 4.4-18 The voltages v_2 , v_3 , and v_4 for the circuit shown in Figure P 4.4-18 are:

$$v_2 = 16 \text{ V}, v_3 = 8 \text{ V}, \text{ and } v_4 = 6 \text{ V}$$

Determine the values of the following:

- The gain, A , of the VCVS
- The resistance R_5
- The currents i_b and i_c
- The power received by resistor R_4

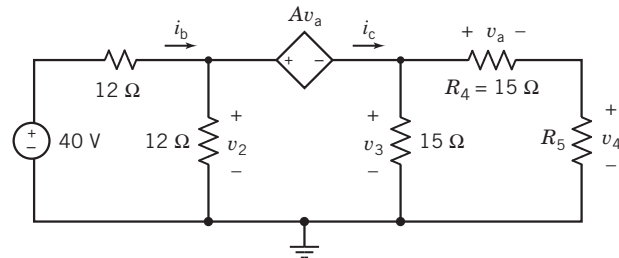


Figure P 4.4-18

P 4.4-19 Determine the values of the node voltages v_1 and v_2 for the circuit shown in Figure P 4.4-19.

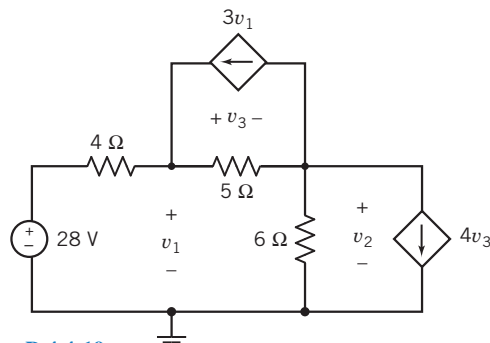


Figure P 4.4-19

P 4.4-20 The encircled numbers in Figure P 4.4-20 are node numbers. Determine the values of v_1 , v_2 , and v_3 , the node voltages corresponding to nodes 1, 2, and 3.

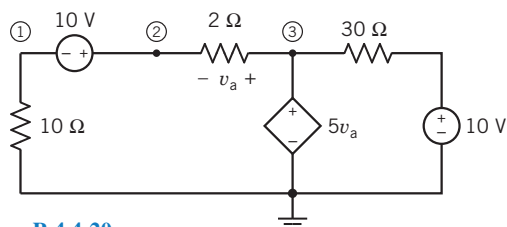


Figure P 4.4-20

P 4.4-21 Determine the values of the node voltages v_1 , v_2 , and v_3 for the circuit shown in Figure P 4.4-21.

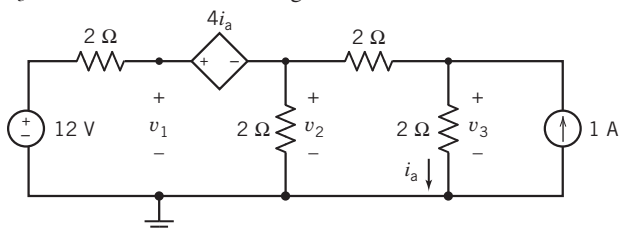


Figure P 4.4-21

P 4.4-22 Determine the values of the node voltages v_1 , v_2 , and v_3 for the circuit shown in Figure P 4.4-22.

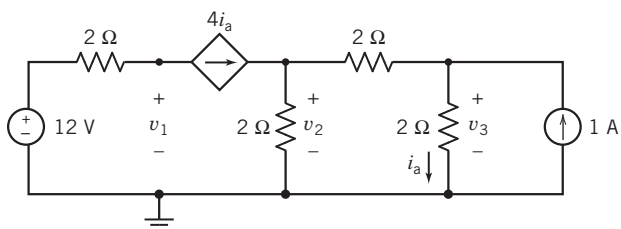


Figure P 4.4-22

Section 4.5 Mesh Current Analysis with Independent Voltage Sources

P 4.5-1 Determine the mesh currents i_1 , i_2 , and i_3 for the circuit shown in Figure P 4.5-1.

Answers: $i_1 = 3$ A, $i_2 = 2$ A, and $i_3 = 4$ A

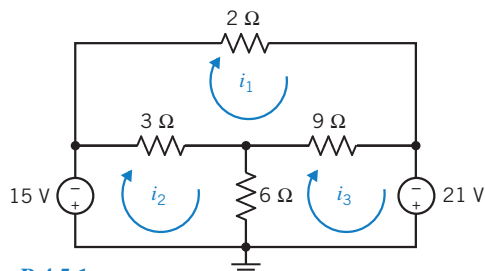


Figure P 4.5-1

P 4.5-2 The values of the mesh currents in the circuit shown in Figure P 4.5-2 are $i_1 = 2$ A, $i_2 = 3$ A, and $i_3 = 4$ A. Determine the values of the resistance R and of the voltages v_1 and v_2 of the voltage sources.

Answers: $R = 12$ Ω, $v_1 = -4$ V, and $v_2 = -28$ V

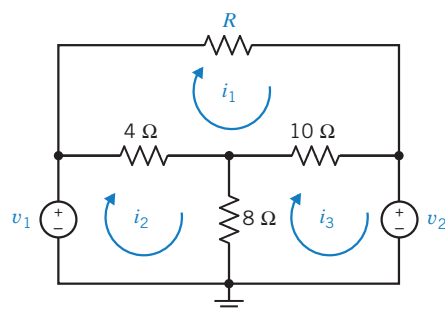


Figure P 4.5-2

P 4.5-3 The currents i_1 and i_2 in Figure P 4.5-3 are the mesh currents. Determine the value of the resistance R required to cause $v_a = -6$ V.

Answer: $R = 4$ Ω

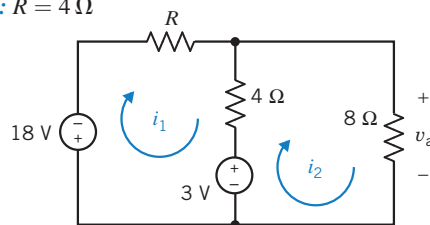


Figure P 4.5-3

P 4.5-4 Determine the mesh currents i_a and i_b in the circuit shown in Figure P 4.5-4.

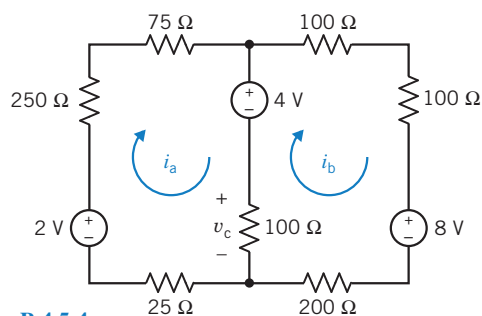


Figure P 4.5-4

P 4.5-5 Find the current i for the circuit of Figure P 4.5-5.

Hint: A short circuit can be treated as a 0-V voltage source.

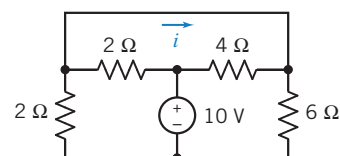


Figure P 4.5-5

P 4.5-6 Simplify the circuit shown in Figure P 4.5-6 by replacing series and parallel resistors by equivalent resistors. Next, analyze the simplified circuit by writing and solving mesh equations.

- (a) Determine the power supplied by each source,
 (b) Determine the power absorbed by the $30\text{-}\Omega$ resistor.

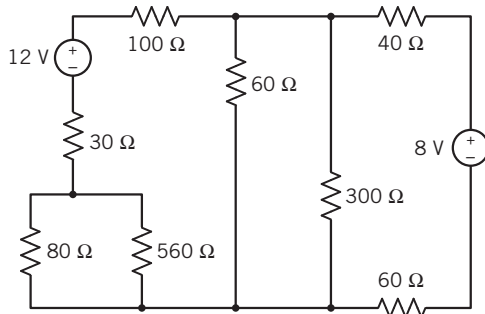


Figure P 4.5-6

Section 4.6 Mesh Current Analysis with Current and Voltage Sources

P 4.6-1 Find i_b for the circuit shown in Figure P 4.6-1.

Answer: $i_b = 0.6\text{ A}$

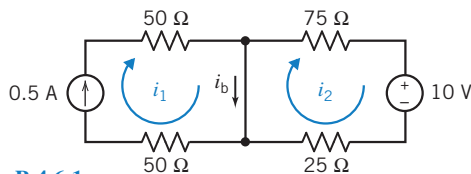


Figure P 4.6-1

P 4.6-2 Find v_c for the circuit shown in Figure P 4.6-2.

Answer: $v_c = 15\text{ V}$

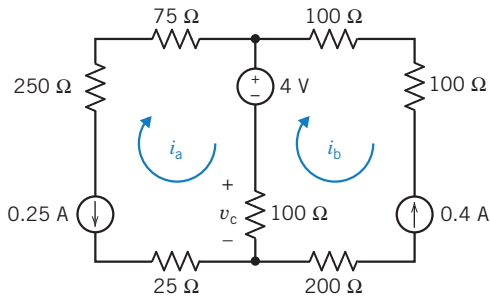


Figure P 4.6-2

P 4.6-3 Find v_2 for the circuit shown in Figure P 4.6-3.

Answer: $v_2 = 2\text{ V}$

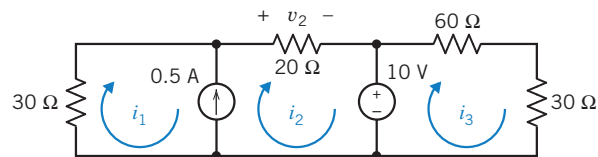


Figure P 4.6-3

P 4.6-4 Find v_c for the circuit shown in Figure P 4.6-4.

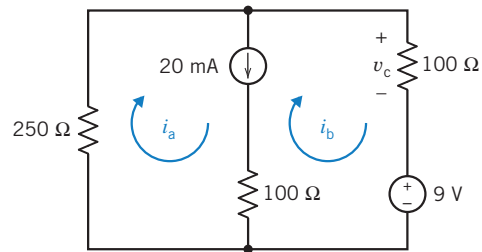


Figure P 4.6-4

P 4.6-5 Determine the value of the voltage measured by the voltmeter in Figure P 4.6-5.

Answer: 8 V

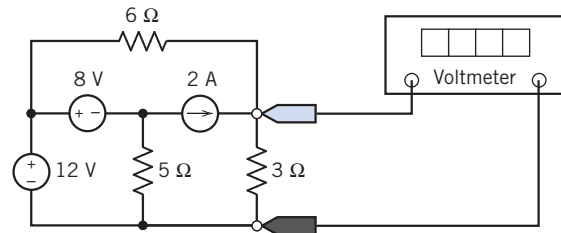


Figure P 4.6-5

P 4.6-6 Determine the value of the current measured by the ammeter in Figure P 4.6-6.

Hint: Write and solve a single mesh equation.

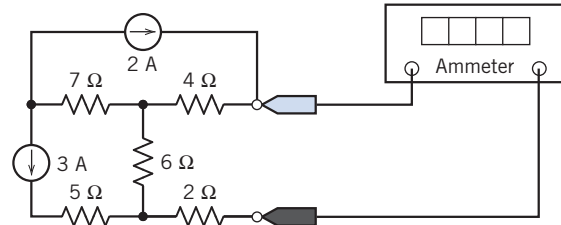


Figure P 4.6-6

P 4.6-7 The mesh currents are labeled in the circuit shown in Figure P 4.6-7. The values of these mesh currents are:

$$i_1 = -1.1014\text{ A}, i_2 = 0.8986\text{ A} \text{ and } i_3 = -0.2899\text{ A}$$

- (a) Determine the values of the resistances R_1 and R_3 .
 (b) Determine the value of the current source current.
 (c) Determine the value of the power supplied by the 12-V voltage source.

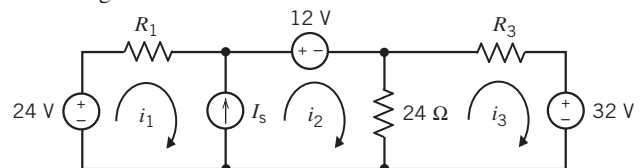


Figure P 4.6-7

P 4.6-8 Determine values of the mesh currents i_1 , i_2 , and i_3 in the circuit shown in Figure P 4.6-8.

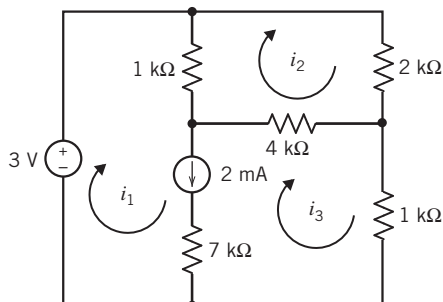


Figure P 4.6-8

P 4.6-9 The mesh currents are labeled in the circuit shown in Figure P 4.6-9. Determine the value of the mesh currents i_1 , and i_2 .

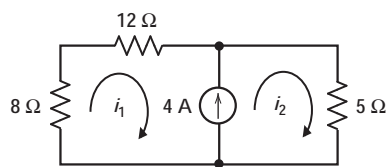


Figure P 4.6-9

P 4.6-10 $\oplus$ The mesh currents in the circuit shown in Figure P 4.6-10 are

$$i_1 = -2.2213 \text{ A}, i_2 = 0.7787 \text{ A}, \text{ and } i_3 = 0.0770 \text{ A}$$

- Determine the values of the resistances R_1 and R_3 .
- Determine the value of the power supplied by the current source.

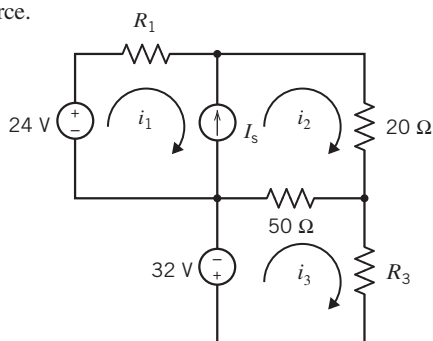


Figure P 4.6-10

P 4.6-11 $\oplus$ Determine the value of the voltage measured by the voltmeter in Figure P 4.6-11.

Hint: Apply KVL to a supermesh to determine the current in the 2-Ω resistor.

Answer: 4/3 V

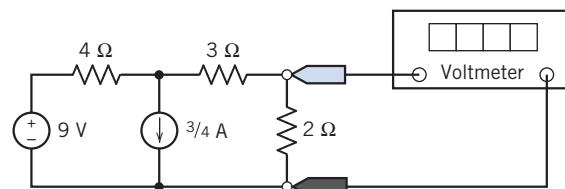


Figure P 4.6-11

P 4.6-12 $\oplus$ Determine the value of the current measured by the ammeter in Figure P 4.6-12.

Hint: Apply KVL to a supermesh.

Answer: -0.333 A

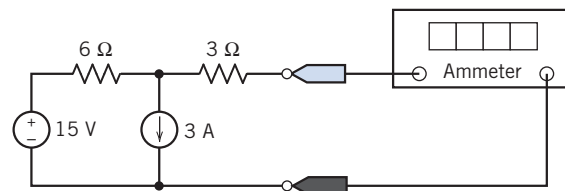


Figure P 4.6-12

P 4.6-13 Determine the values of the mesh currents i_1 , i_2 , and i_3 and the output voltage v_o in the circuit shown in Figure P 4.6-13.

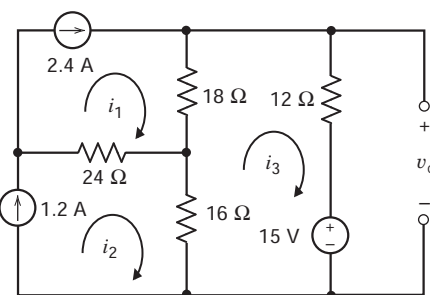


Figure P 4.6-13

P 4.6-14 Determine the values of the power supplied by the sources in the circuit shown in Figure P 4.6-14.

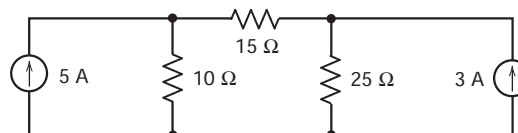


Figure P 4.6-14

P 4.6-15 Determine the values of the resistance R and of the power supplied by the 6-A current source in the circuit shown in Figure P 4.6-15.

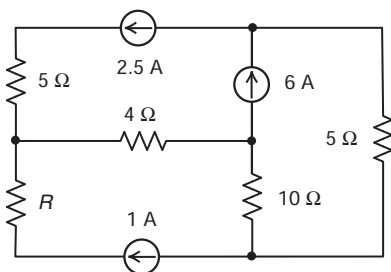


Figure P 4.6-15

Section 4.7 Mesh Current Analysis with Dependent Sources

P 4.7-1 Find v_2 for the circuit shown in Figure P 4.7-1.

Answer: $v_2 = 10$ V

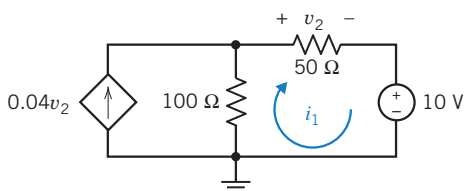


Figure P 4.7-1

P 4.7-2 Determine the values of the power supplied by the voltage source and by the CCCS in the circuit shown in Figure P 4.7-2.

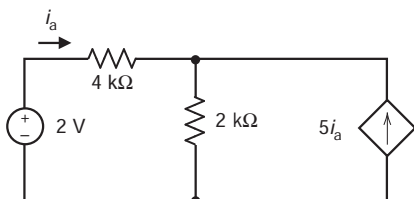


Figure P 4.7-2

P 4.7-3 Find v_o for the circuit shown in Figure P 4.7-3.

Answer: $v_o = 2.5$ V

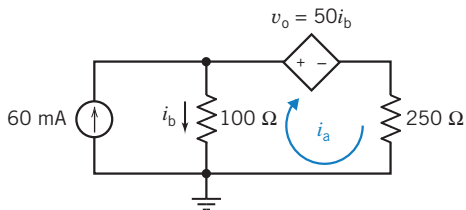


Figure P 4.7-3

P 4.7-4 Determine the mesh current i_a for the circuit shown in Figure P 4.7-4.

Answer: $i_a = -24$ mA

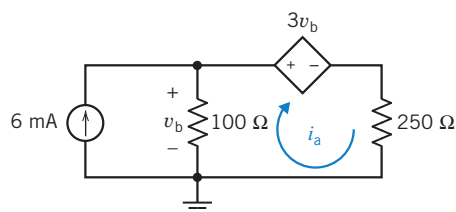


Figure P 4.7-4

P 4.7-5 Although scientists continue to debate exactly why and how it works, the process of using electricity to aid in the repair and growth of bones—which has been used mainly with fractures—may soon be extended to an array of other problems, ranging from osteoporosis and osteoarthritis to spinal fusions and skin ulcers.

An electric current is applied to bone fractures that have not healed in the normal period of time. The process seeks to imitate natural electrical forces within the body. It takes only a small amount of electric stimulation to accelerate bone recovery. The direct current method uses an electrode that is implanted at the bone. This method has a success rate approaching 80 percent.

The implant is shown in Figure P 4.7-5a, and the circuit model is shown in Figure P 4.7-5b. Find the energy delivered to the cathode during a 24-hour period. The cathode is represented by the dependent voltage source and the 100-kΩ resistor.

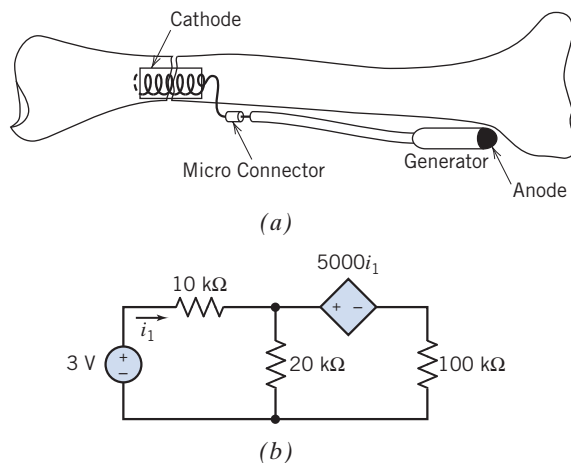


Figure P 4.7-5 (a) Electric aid to bone repair. (b) Circuit model.

P 4.7-6 Determine the value of the power supplied by the VCCS in the circuit shown in Figure P 4.7-6.

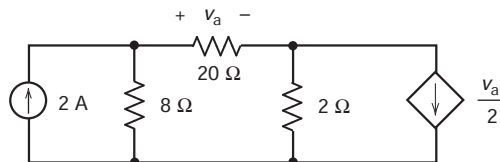


Figure P 4.7-6

P 4.7-7 The currents i_1 , i_2 , and i_3 are the mesh currents of the circuit shown in Figure P 4.7-7. Determine the values of i_1 , i_2 , and i_3 .

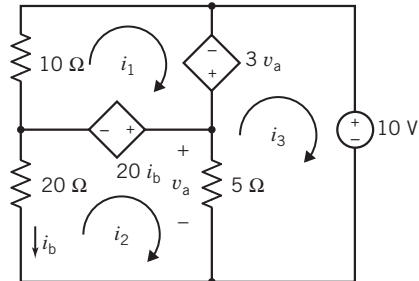


Figure P 4.7-7

P 4.7-8 $\oplus$ Determine the value of the power supplied by the dependent source in Figure P 4.7-8.

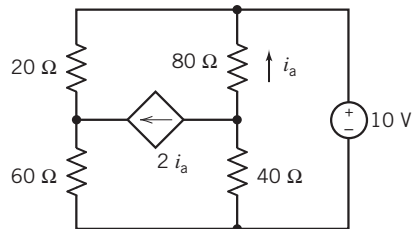


Figure P 4.7-8

P 4.7-9 $\oplus$ Determine the value of the resistance R in the circuit shown in Figure P 4.7-9.

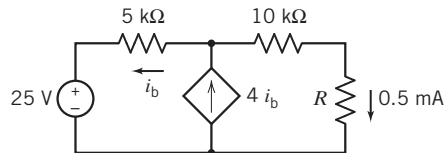


Figure P 4.7-9

P 4.7-10 The circuit shown in Figure P 4.7-10 is the small signal model of an amplifier. The input to the amplifier is the voltage source voltage v_s . The output of the amplifier is the voltage v_o .

- The ratio of the output to the input, v_o/v_s , is called the gain of the amplifier. Determine the gain of the amplifier.
- The ratio of the current of the input source to the input voltage i_b/v_s is called the input resistance of the amplifier. Determine the input resistance.

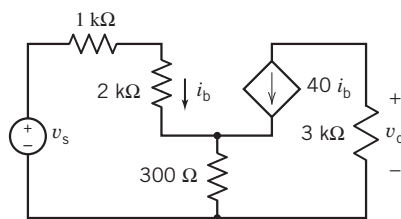


Figure P 4.7-10

P 4.7-11 Determine the values of the mesh currents of the circuit shown in Figure P 4.7-11.

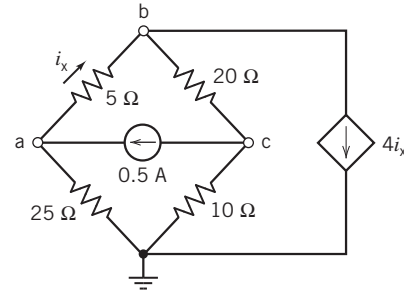


Figure P 4.7-11

P 4.7-12 The currents i_1 , i_2 , and i_3 are the mesh currents corresponding to meshes 1, 2, and 3 in Figure P 4.7-12. Determine the values of these mesh currents.

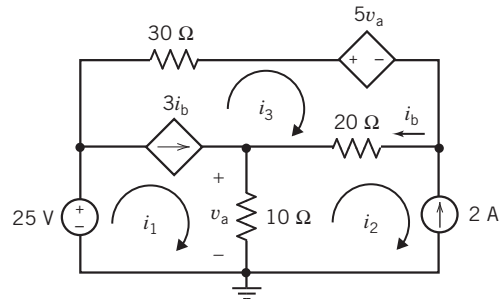


Figure P 4.7-12

P 4.7-13 The currents i_1 , i_2 , and i_3 are the mesh currents corresponding to meshes 1, 2, and 3 in Figure P 4.7-13. The values of these currents are

$$i_1 = -1.375 \text{ A}, i_2 = -2.5 \text{ A} \text{ and } i_3 = -3.25 \text{ A}$$

Determine the values of the gains of the dependent sources, A and B .

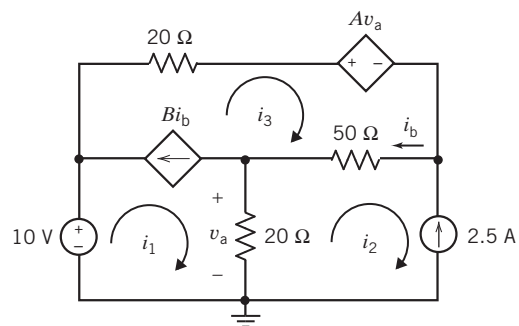


Figure P 4.7-13

P 4.7-14 Determine the current i in the circuit shown in Figure P 4.7-14.

Answer: $i = 3 \text{ A}$

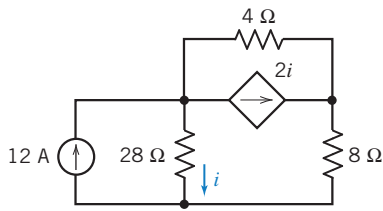


Figure P 4.7-14

P 4.7-15 Determine the values of the mesh currents i_1 and i_2 for the circuit shown in Figure P 4.7-15.

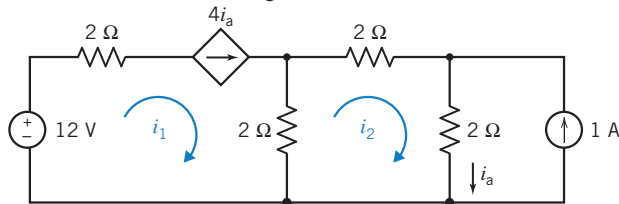


Figure P 4.7-15

P 4.7-16 Determine the values of the mesh currents i_1 and i_2 for the circuit shown in Figure P 4.7-16.

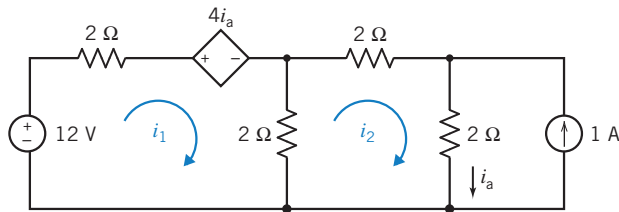


Figure P 4.7-16

Section 4.8 The Node Voltage Method and Mesh Current Method Compared

P 4.8-1 The circuit shown in Figure P 4.8-1 has two inputs, v_s and i_s , and one output, v_o . The output is related to the inputs by the equation

$$v_o = ai_s + bv_s$$

where a and b are constants to be determined. Determine the values a and b by (a) writing and solving mesh equations and (b) writing and solving node equations.

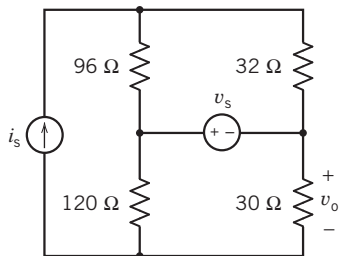


Figure P 4.8-1

P 4.8-2 Determine the power supplied by the dependent source in the circuit shown in Figure P 4.8-2 by writing and solving (a) node equations and (b) mesh equations.

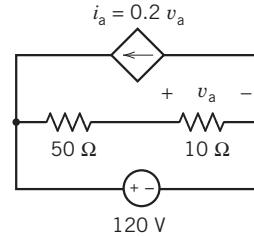


Figure P 4.8-2

Section 4.9 Circuit Analysis Using MATLAB

P 4.9-1 The encircled numbers in the circuit shown Figure P 4.9-1 are node numbers. Determine the values of the corresponding node voltages v_1 , v_2 , and v_3 .

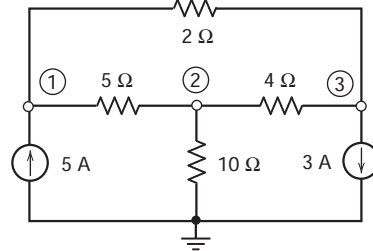


Figure P 4.9-1

P 4.9-2 Determine the values of the node voltages v_1 and v_2 in the circuit shown in Figure P 4.9-2.

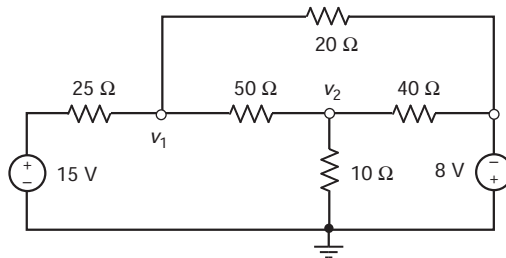


Figure P 4.9-2

P 4.9-3 Determine the values of the node voltages v_1 , v_2 , and v_3 in the circuit shown in Figure P 4.9-3.

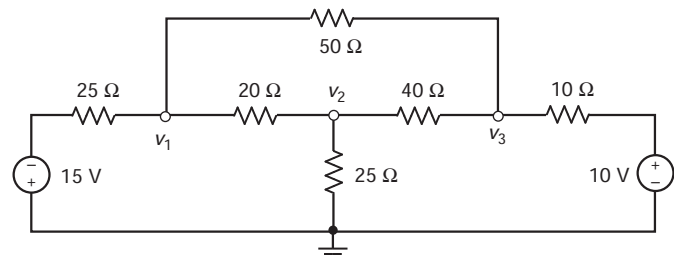


Figure P 4.9-3

P 4.9-4 Determine the node voltages v_1 and v_2 for the circuit shown in Figure P 4.9-4.

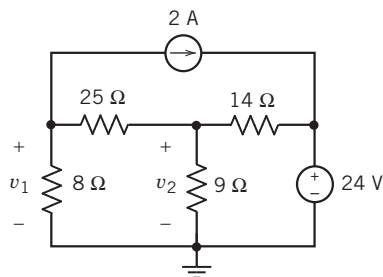


Figure P 4.9-4

P 4.9-5 Determine the mesh currents i_1 and i_2 for the circuit shown in Figure P 4.9-5.

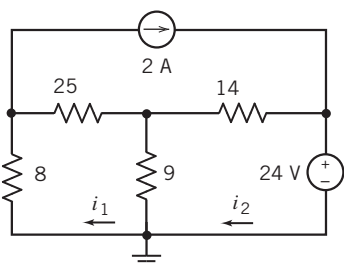


Figure P 4.9-5

P 4.9-6 Represent the circuit shown in Figure P 4.9-6 by the matrix equation

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} -40 \\ -228 \end{bmatrix}$$

Determine the values of the coefficients a_{11} , a_{12} , a_{21} , and a_{22} .

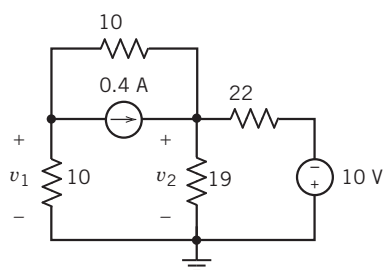


Figure P 4.9-6

P 4.9-7 Represent the circuit shown in Figure P 4.9-7 by the matrix equation

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \end{bmatrix} = \begin{bmatrix} 4 \\ 10 \end{bmatrix}$$

Determine the values of the coefficients a_{11} , a_{12} , a_{21} , and a_{22} .

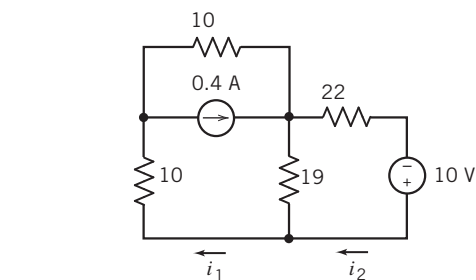


Figure P 4.9-7

P 4.9-8 Determine the values of the power supplied by each of the sources for the circuit shown in Figure P 4.9-8.

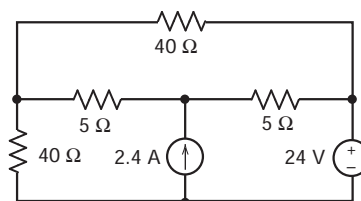


Figure P 4.9-8

P 4.9-9 The mesh currents are labeled in the circuit shown in Figure P 4.9-9. Determine the value of the mesh currents i_1 and i_2 .

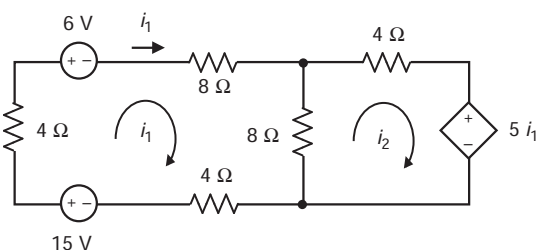


Figure P 4.9-9

P 4.9-10 The encircled numbers in the circuit shown in Figure P 4.9-10 are node numbers. Determine the values of the corresponding node voltages v_1 and v_2 .

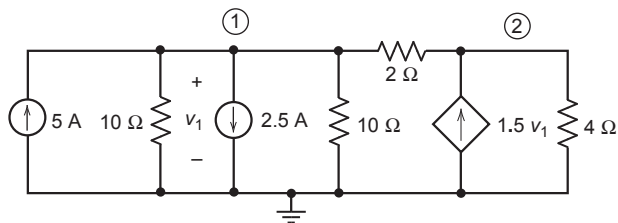


Figure P 4.9-10

Section 4.11 How Can We Check . . . ?

P 4.11-1 Computer analysis of the circuit shown in Figure P 4.11-1 indicates that the node voltages are $v_a = 5.2$ V, $v_b = -4.8$ V, and $v_c = 3.0$ V. Is this analysis correct?

Hint: Use the node voltages to calculate all the element currents. Check to see that KCL is satisfied at each node.

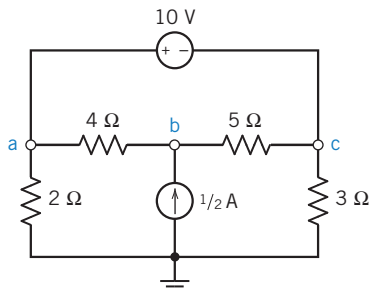


Figure P 4.11-1

P 4.11-2 $\oplus$ An old lab report asserts that the node voltages of the circuit of Figure P 4.11-2 are $v_a = 4$ V, $v_b = 20$ V, and $v_c = 12$ V. Are these correct?

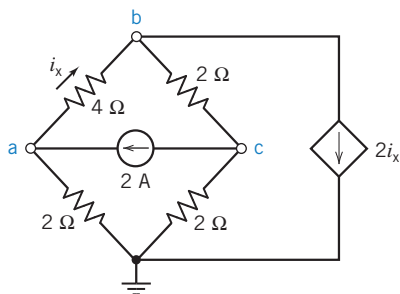


Figure P 4.11-2

P 4.11-3 Your lab partner forgot to record the values of R_1 , R_2 , and R_3 . He thinks that two of the resistors in Figure P 4.11-3 had values of 10 k Ω and that the other had a value of 5 k Ω . Is this possible? Which resistor is the 5-k Ω resistor?

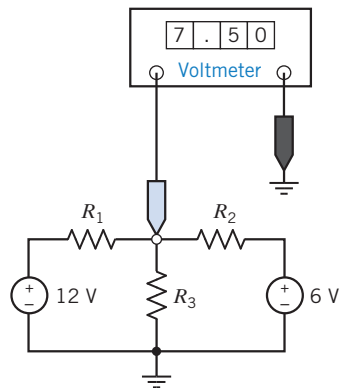


Figure P 4.11-3

P 4.11-4 Computer analysis of the circuit shown in Figure P 4.11-4 indicates that the mesh currents are $i_1 = 2$ A, $i_2 = 4$ A, and $i_3 = 3$ A. Verify that this analysis is correct.

Hint: Use the mesh currents to calculate the element voltages. Verify that KVL is satisfied for each mesh.

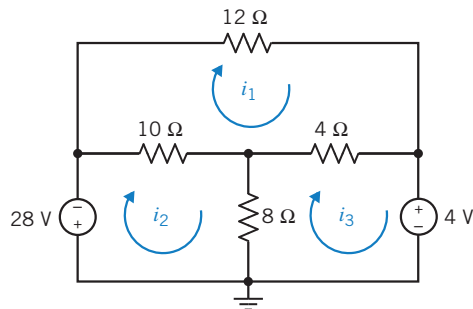


Figure P 4.11-4